Coherent Summary and Cross-Session Connections
Optimization → Demand & Duality → Equilibrium & Welfare → Empirical Rationality

Global Spine: One Coherent Line Across All Sessions

All sessions fit into one pipeline:

$$\boxed{\textbf{Optimization as a behavioral model}} \Rightarrow \boxed{\textbf{Demand and value functions}} \Rightarrow$$

$$\Rightarrow \boxed{\textbf{Comparative statics and welfare}} \Rightarrow$$

$$\Rightarrow \boxed{\textbf{Aggregation and equilibrium}} \Rightarrow \boxed{\textbf{Testability from data (revealed preference)}}$$

- **Optimization** gives a disciplined way to generate predictions: choices solve constrained problems.
- **Demand and value functions** ($x(p, m)$, $h(p, u)$, $v(p, m)$, $e(p, u)$) compress behavior into objects that can be differentiated, compared, aggregated, and welfare-evaluated.
- **Comparative statics** (e.g., Slutsky decomposition) explains *mechanisms* behind changes in behavior.
- **Equilibrium** adds market feasibility (clearing) and production; welfare theorems connect decentralized equilibrium to efficiency.
- **Revealed preference** asks the reverse question: given observed choices, is there *any* utility model consistent with optimization?

Key unifier: the same toolkit appears repeatedly under different names:

$$\text{KKT / Lagrange} \quad + \quad \text{duality} \quad + \quad \text{envelope logic} \quad + \quad \text{market feasibility.}$$

Module A: Consumer Choice and KKT (Utility Maximization)

A.1 Core objects

- Utility maximization (UMP):

$$\max_{x \geq 0} u(x) \quad \text{s.t.} \quad p \cdot x \leq m.$$

- Outputs: Marshallian demand $x(p, m)$ and indirect utility $v(p, m) = u(x(p, m))$.

A.2 Why KKT is the universal solver

- Inequality constraints (budget, nonnegativity) lead naturally to KKT conditions.
- The economic interpretation of KKT is structural: which constraints bind determines whether the solution is interior, corner, or kink.

A.3 Connection forward

This module provides the **behavioral foundation** for everything later:

- To study welfare and policy, we need not only $x(p, m)$ but also value functions $v(p, m)$.
- To explain mechanisms (substitution vs income), we need the dual objects from expenditure minimization.
- For equilibrium, consumers must solve UMP given prices.

Module B: Expenditure Minimization, Duality, and Comparative Statics

B.1 The dual problem and dual objects

- Expenditure minimization (EMP):

$$e(p,\bar{u}) = \min_{x \geq 0} \; p \cdot x \quad \text{s.t.} \quad u(x) \geq \bar{u}.$$

- Outputs: Hicksian demand $h(p,\bar{u})$ and expenditure function $e(p,\bar{u})$.

B.2 Duality: the bridge between "money" and "utility"

Duality identities link the two worlds:

$$e\big(p, v(p,m)\big) = m, \qquad v\big(p, e(p,\bar{u})\big) = \bar{u},$$

$$x(p,m) = h\big(p, v(p,m)\big), \qquad h(p,\bar{u}) = x\big(p, e(p,\bar{u})\big).$$

B.3 Envelope logic as a reusable principle

Differentiating value functions yields behavior:

- Shephard-type logic: derivatives of the cost/expenditure function return compensated demands.
- Roy-type logic: derivatives of the indirect utility return Marshallian demands.

B.4 Mechanisms: Slutsky decomposition

$$\frac{\partial x_i(p,m)}{\partial p_j} = \underbrace{\frac{\partial h_i(p,\bar{u})}{\partial p_j}}_{\text{substitution effect}} - \underbrace{\frac{\partial x_i(p,m)}{\partial m} x_j(p,m)}_{\text{income effect}}, \qquad \bar{u} = v(p,m).$$

B.5 Connection forward

This module is the **mechanism and welfare engine**:

- It explains *why* behavior changes (substitution vs income).
- It provides the objects used later for welfare evaluation (EV/CV are written using $e(\cdot)$).
- It also prepares aggregation results (via properties of indirect utility and demand).

Module C: Production, Firms, and Duality on the Supply Side

C.1 Producer problems

- Profit maximization (PMP):

$$\max_{y \in Y} p \cdot y, \quad \text{profit function } \pi(p).$$

- Cost minimization (CMP):

$$C(w, q) = \min_{z \geq 0} \; w \cdot z \quad \text{s.t.} \quad f(z) \geq q.$$

C.2 Supply-side duality and envelope logic

- PMP and CMP are dual descriptions of firm behavior (choose q to maximize $pq - C(w, q)$ once C is known).
- Profit/cost functions inherit homogeneity and curvature properties; differentiating them recovers optimal supplies and factor demands.

C.3 Connection forward

This module supplies the **other half of equilibrium**:

- In general equilibrium, producers solve PMP as consumers solve UMP.
- Curvature/homogeneity properties are important for existence, comparative statics, and welfare results.

Module D: Competitive Equilibrium and Welfare Theorems

D.1 Equilibrium logic

A competitive equilibrium is defined by:

- Consumer optimality (UMP at given prices),
- Firm optimality (PMP at given prices),
- Market clearing (aggregate demand equals aggregate endowment plus net production).

D.2 Efficiency viewpoint

- Pareto efficiency is defined by the impossibility of a feasible Pareto improvement.
- A planner problem with welfare weights characterizes the Pareto set.

D.3 Welfare theorems as "connection results"

- First welfare theorem: equilibrium $\Rightarrow$ Pareto efficient allocation (under standard assumptions).
- Second welfare theorem: any Pareto efficient allocation can be decentralized as an equilibrium after appropriate redistribution.

D.4 Connection forward

This module explains **why micro-foundations matter**:

- Individual optimization + feasibility constraints produce market-level outcomes.
- The welfare theorems motivate welfare measurement (EV/CV) and redistribution analysis.

Session 1 (Technical Foundation): When KKT Works and When It Fails

1. Why this session exists (the meta-message)

Earlier modules used KKT as a solver. This session clarifies the **logical status** of KKT:

- When is "KKT $\Rightarrow$ optimal" valid?
- When can KKT generate false candidates?
- When can an optimum exist but KKT fail to describe it?

2. Two cautionary facts

- **KKT points may not be solutions:** non-concave problems can have KKT points that are only local or even non-optimal.
- **Solutions may not be KKT points:** if constraint qualification fails, the standard KKT system need not characterize the optimum.

3. The key sufficient condition: concave programming

If the objective and constraints form a concave program, then **any KKT point is a global optimum**. This is the main reason curvature (concavity/convexity) is emphasized throughout.

4. Equality constraints and constraint qualification

- Equality constraints can be represented as paired inequalities, leading to an unrestricted multiplier.
- With equality constraints, concave programming requires the equality constraint to be linear.
- Constraint qualification (e.g., linear independence of active gradients) explains when KKT is necessary for optimality.

5. Connection to the rest (why it matters)

This session is the **validity check** for the entire framework:

- Consumer/producer derivations rely on KKT—this session tells you when those derivations are logically safe.
- Curvature assumptions used in demand/supply properties and welfare results are exactly the assumptions that make KKT reliable.

Session 2 (Empirical Foundation): Revealed Preference, Aggregation, and Welfare Measures

1. Revealed preference: reversing the direction of the model

Earlier modules assume a utility function and derive choices. Revealed preference asks:

$$\text{Given observed } (p, m, x), \text{ does there exist a } u(\cdot) \text{ for which } x \text{ is optimal?}$$

This turns micro theory into a **testable discipline**.

2. WARP and SARP as consistency conditions

- WARP rules out pairwise contradictions: if one chosen bundle was affordable under another budget, the reverse strict affordability cannot occur.
- SARP rules out cycles across multiple observations; it is stronger and closer to "existence of a utility representation" for finite datasets.

3. Aggregation and the distribution of wealth

Aggregate demand is:

$$X(p; m_1, \ldots, m_I) = \sum_{i=1}^{I} x_i(p, m_i).$$

In general, X depends on the full distribution $(m_1, \ldots, m_I)$, not only on total wealth. A special condition (Gorman form) yields aggregation that depends only on total wealth.

4. Welfare measures: EV and CV

Price changes can be evaluated in money units using the expenditure function:

- Equivalent variation (EV): money at old prices equivalent to the price change.
- Compensating variation (CV): money at new prices needed to restore the original utility.
- Both are written using $e(p, u)$ and can be expressed as integrals of Hicksian demand (via Shephard-type logic).

5. Connection to the rest (why it matters)

This session completes the circle:

- It uses the optimization framework to build **observable** restrictions (WARP/SARP).
- It uses duality and Hicksian demand to build **money-metric welfare** (EV/CV).
- It clarifies when micro predictions survive aggregation and when distribution matters.

Higher-Level Coherence Summary

1. **Micro begins with optimization.** Consumers and firms solve constrained problems; KKT is the standard language for describing their optimality.
2. **Duality turns behavior into tractable objects.** The four objects

$$x(p,m),\; h(p,u),\; v(p,m),\; e(p,u)$$

   let you move between choices and values, and between money and utility.
3. **Mechanisms come from derivatives of value functions.** Envelope logic yields Shephard/Roy-style relationships; Slutsky decomposes price effects into substitution and income.
4. **Markets aggregate individuals.** Equilibrium adds feasibility (clearing) and production; welfare theorems connect equilibrium to efficiency and redistribution.
5. **Theory becomes testable and policy-relevant.** Revealed preference turns optimization into restrictions on data; EV/CV convert welfare into dollars.

One-line takeaway: micro is a single coherent system that turns *preferences and technology* into *choices*, explains *mechanisms* behind changes, aggregates them into *market outcomes*, and then evaluates those outcomes using *welfare metrics* and *testable restrictions*.